

Embedded and Situated Visualisation in Mixed Reality to Support Interval Running — A Research Highlight

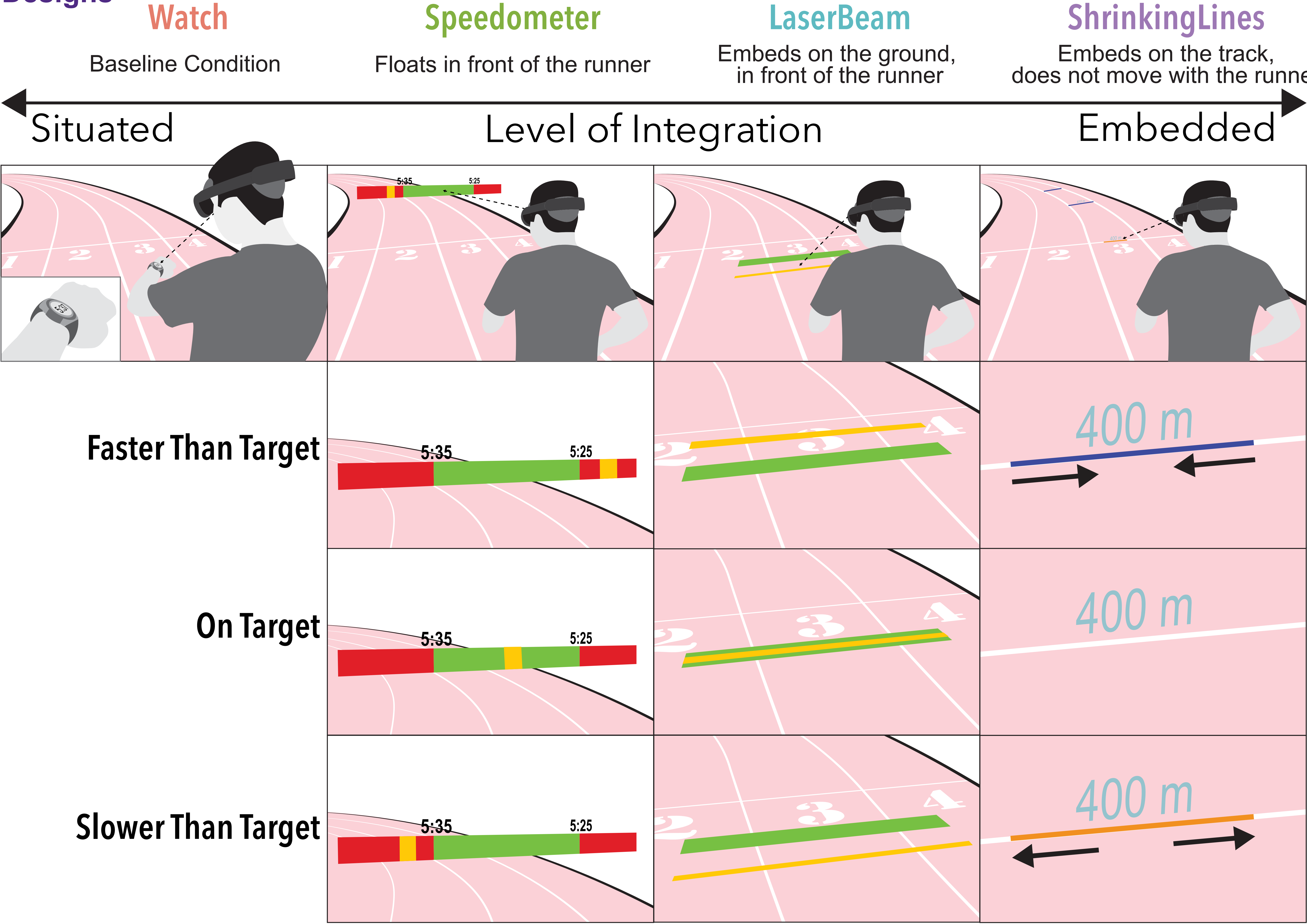
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Introduction

Running is becoming more and more data-driven. However, smartwatches are small and hard to read while running. We investigate the use of mixed reality visualisations to help pace tracking for interval running. We introduce three immersive visual designs to support pace tracking. Our designs leverage the level of integration/embeddedness of MR visualisation to embed pace visualisation in to the surrounding physical environment.

Designs



Evaluation

Apparatus: Outdoor 400m Running Track, Meta Quest Pro Participants: 16 frequent runners, within-subject

Evaluation Task: Interval running, altering pace between high intensity intervals (at 5:30 mins/km for 50s) and low intensity intervals (at 7:30 mins/km for 50s) for 3 times (5 mins per condition and 20 mins in total)

Data Collection and Result

Pace Irregularity: Both **Speedometer** and **LaserBeam** had a smaller pace irregularity when compared with **Watch** and **ShrinkingLines**
Task Workload: Both **LaserBeam** and **ShrinkingLines** required a smaller amount of overall workload than **Watch** and **Speedometer**
Flow State: Participants were more “in the flow” with **LaserBeam** than with **Watch** and than with **Speedometer**, and more “in the flow” with **ShrinkingLines** than with **Watch** and than **Speedometer**
Preference Ranking: Participants clearly preferred **LaserBeam**, then **ShrinkingLines**. They ranked **Speedometer** and **Watch** similarly, although **Watch** was the least preferred overall

Conclusion

All of our immersive designs are at least on par or better than the **Watch** baseline. Embedded visual representations like **LaserBeam** and **ShrinkingLines** enhance both performance and user satisfaction.



Read the paper



Check out the video



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